



Singh-Soluble-20747-7

Description

SOLUBLE (NaCl-Extracted) OVARIAN SAMPLES

Analysis

#1

Report creation date

2026-01-28

Report created with

TARAM v2.11.2

The ZIP file includes two folders:

- **plots:** Contains all plots in PNG, SVG and interactive, clickable HTML formats.
- **tables:** Contains all tables in tab-separated format.

Overview of data analysis

|    |                    |                               |
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Conclusion

Table-17 is the final table. Enrichment-table-13 contains the results of the GO term enrichment.

-----comments-----

Despite some variability in the data for some sample, the differences between time points are strong and clear.

1. Initial result

Table 1

7272 protein groups

Default intensity column:

Quantity

Software version:

Spectronaut 20.4.260109.92449

Analysis date:

24-January-2026

Fasta files:

RefProt\_Mus\_musculus\_20250206.fasta  
Universal\_Contaminant\_Protein\_mod\_20251210.fasta

Groups and sample identities:

| D14   | D21   | M03   | M06   | M12   | M18        |
|-------|-------|-------|-------|-------|------------|
| 20747 | 20752 | 20757 | 20762 | 20768 | 20775b     |
| 20748 | 20753 | 20758 | 20763 | 20769 | 20776-40ul |
| 20749 | 20754 | 20759 | 20764 | 20770 | 20777      |
| 20750 | 20755 | 20760 | 20765 | 20771 | 20778      |
| 20751 | 20756 | 20761 | 20766 | 20772 | 20779      |

Samples in liquid forn were digested by the SP3 method. MS analysis was on the TIMS-TOF instrument by DIA, with EvoSep LC separation (whisper Zoom). Data analysis was done by Spectronaut in directDIA mode, with local normalization (default).

2. Log transformation

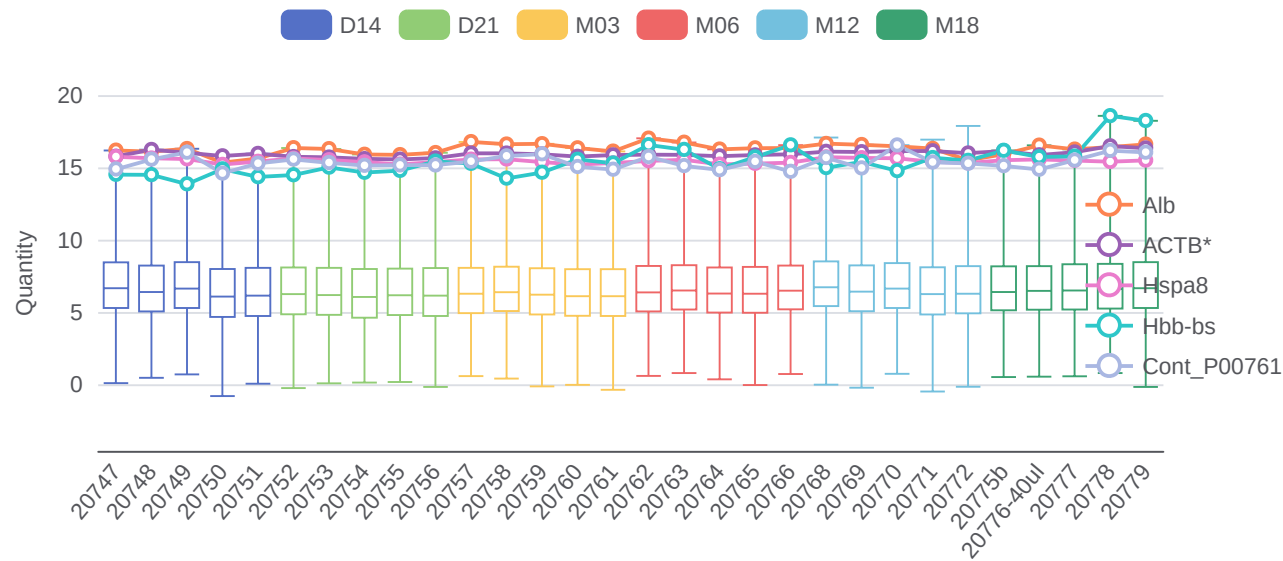
Table 2

7272 protein groups

*Log transformation facilitates plotting of data and makes distributions more “normal”, allowing application of standard statistical tests.*

3. Boxplot

Overview of samples. Y-axis is usually in log2 scale.



\* only the first of multiple gene names is displayed.

In a majority of samples, albumin was the most abundant component. Hemoglobin was also found in high concentration.

4. Filter rows

Table 4

7182 protein groups

Parameters:  
Remove potential contaminants  
Remove `PG.ProteinGroups == *Cont_*`

Protein groups removed: 90

## 5. Summary

"Nr of valid" = number of quantified proteins.

"NaN" (Not a Number) = missing values.

"Nr of peptides" = sum of "NrOfPrecursorsIdentified"

| Name       | Group | Min   | Max   | Median | Nr of valid | Nr of NaN | Nr of peptides |
|------------|-------|-------|-------|--------|-------------|-----------|----------------|
| 20747      | D14   | 0.07  | 16.17 | 6.63   | 5346        | 1836      | 64352          |
| 20748      |       | 0.43  | 16.09 | 6.37   | 5706        | 1476      | 72288          |
| 20749      |       | 0.67  | 16.29 | 6.6    | 5329        | 1853      | 61363          |
| 20750      |       | -0.83 | 15.32 | 6.05   | 6488        | 694       | 92307          |
| 20751      |       | 0.02  | 15.62 | 6.11   | 6164        | 1018      | 83462          |
| 20752      | D21   | -0.28 | 16.34 | 6.22   | 5933        | 1249      | 73345          |
| 20753      |       | 0.05  | 16.28 | 6.16   | 6139        | 1043      | 81043          |
| 20754      |       | 0.1   | 15.89 | 6.02   | 6291        | 891       | 89353          |
| 20755      |       | 0.14  | 15.87 | 6.15   | 6415        | 767       | 89973          |
| 20756      |       | -0.2  | 16.01 | 6.12   | 6267        | 915       | 87528          |
| 20757      | M03   | 0.55  | 16.77 | 6.25   | 6080        | 1102      | 75962          |
| 20758      |       | 0.39  | 16.6  | 6.35   | 5850        | 1332      | 67218          |
| 20759      |       | -0.16 | 16.62 | 6.18   | 6049        | 1133      | 74390          |
| 20760      |       | -0.05 | 16.33 | 6.08   | 6405        | 777       | 87084          |
| 20761      |       | -0.4  | 16.11 | 6.07   | 6347        | 835       | 84539          |
| 20762      | M06   | 0.56  | 17.01 | 6.34   | 5823        | 1359      | 71135          |
| 20763      |       | 0.76  | 16.72 | 6.47   | 5645        | 1537      | 65807          |
| 20764      |       | 0.33  | 16.25 | 6.27   | 6174        | 1008      | 76077          |
| 20765      |       | -0.06 | 16.33 | 6.24   | 6027        | 1155      | 72303          |
| 20766      |       | 0.69  | 16.54 | 6.45   | 5724        | 1458      | 67566          |
| 20768      | M12   | -0.03 | 17.07 | 6.69   | 5200        | 1982      | 60195          |
| 20769      |       | -0.25 | 16.56 | 6.38   | 5888        | 1294      | 69522          |
| 20770      |       | 0.71  | 16.46 | 6.61   | 5379        | 1803      | 62417          |
| 20771      |       | -0.51 | 16.92 | 6.21   | 6021        | 1161      | 75666          |
| 20772      |       | -0.19 | 17.87 | 6.24   | 6086        | 1096      | 78500          |
| 20775b     | M18   | 0.48  | 16.16 | 6.37   | 5963        | 1219      | 69389          |
| 20776-40ul |       | 0.52  | 16.52 | 6.43   | 5905        | 1277      | 66617          |
| 20777      |       | 0.54  | 16.25 | 6.48   | 5631        | 1551      | 69563          |
| 20778      |       | 0.76  | 18.56 | 6.58   | 5620        | 1562      | 72083          |
| 20779      |       | -0.2  | 18.23 | 6.62   | 5422        | 1760      | 63687          |

Numbers of ID's varied quite a bit (+/- 25%). This is not unexpected, since in purified samples a similar variation is often observed. The absolute number of identifications is also somehow impacted by the presence of a few abundant species like albumin.

6. Filter on valid

Table 6

Retain only proteins quantified in a minimum number of samples.

6599 protein groups

Parameters:Protein groups removed: 583

Only keep rows where *Quantity* has at least 4 valid value(s) in one group.

7. Filter rows

Table 7

6286 protein groups

Parameters:Protein groups removed: 313

Only keep *PG.NrOfPrecursorsIdentified (Experiment-wide)* > 1

8. Imputation

Table 8

Imputation allows calculating fold changes and statistical tests for proteins with missing values. NaN (Not a Number) values are replaced by low-shifted random values, based on the assumption that missing values occur when signals are below detection limits.

6286 protein groups

Parameters:Nr of imputed values: 20757

Replace missing values in column(s) [Quantity...] by random numbers drawn from a normal distribution:Nr of protein groups with imputation: 2536

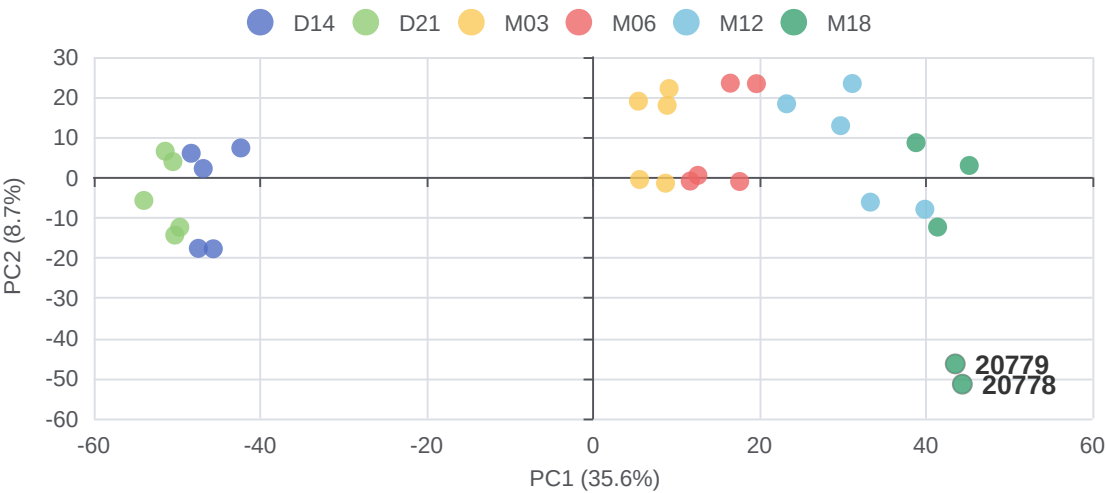
Width: 0.3

Downshift: 1.8

Seed: 1

9. PCA

Examines similarities between samples and groups, needs imputed values.



Separation of groups is quite impressive, considering the somewhat varying numbers of ID's. Kinetics can be followed beautifully. Samples 20778-79 appear like outliers in PC2

10. t-test

Table 10

Benjamini-Hochberg correction adjusts thresholds for large datasets.

6286 protein groups

Welch's t-test (assumes unequal variances)

Significance threshold: 0.05

Multiple testing correction: Benjamini-Hochberg (FDR)

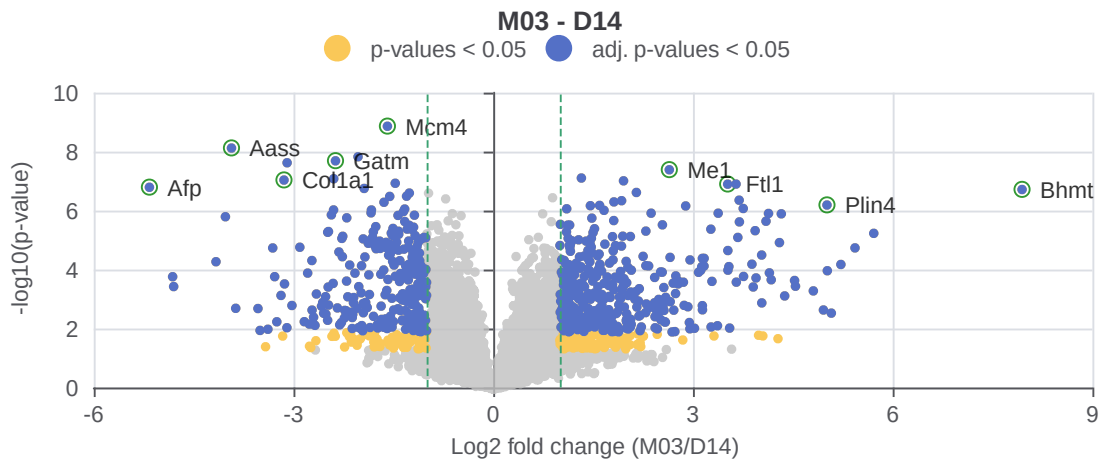
Only compute comparisons when there are at least 4 valid (non-imputed) values in one group.

| Comparison | Significant | Filter passed |
|------------|-------------|---------------|
| M03 - D14  | 1581        | 5907          |
| M18 - M03  | 415         | 5830          |

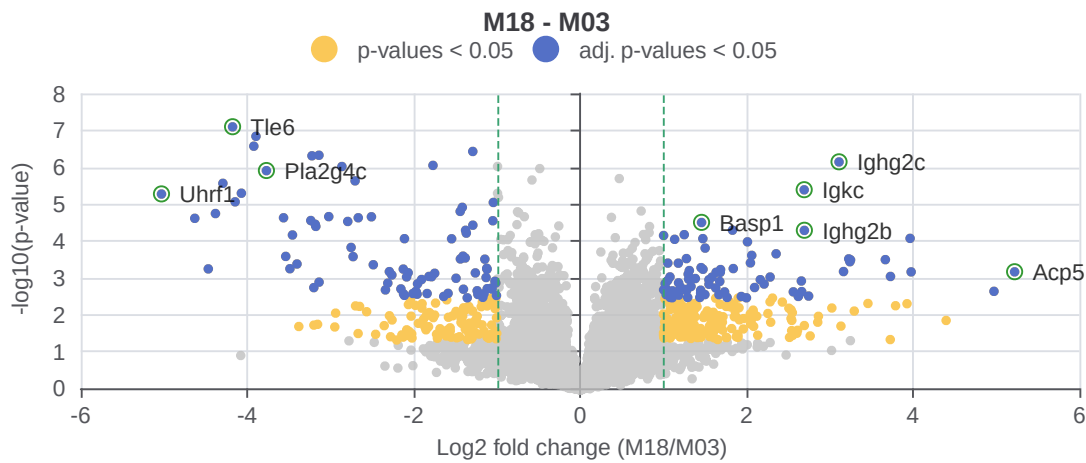
Only 2 major comparisons done here

11. Volcano plot

Common way to represent statistical significance (y-axis) and fold change (x-axis). By using  $-\log_{10}$  transformation, low  $p$ -values are displayed at the top of the plot.



12. Volcano plot



Immunoglobulins upregulated in M18 samples

### 13. 1D Enrichment

Table 13

Table is first ranked by (usually) fold change. Distribution of each annotation term is then evaluated to see if it is enriched toward the top or bottom. FDR-filtered table.

Enrichment table 13

6286 protein groups

**Parameters:**  
**Annotation file:** mainAnnot.mus\_musculus.txt  
**Annotation info:** Mus musculus, PerseusAnnotation from 18-12-23, 2023-12-18, 86221 entries.  
**Selected annotations:** GOMF name, GOBP slim name, GOCC slim name, KEGG name  
**Selected column(s):** t.statistic.M03-D14; t.statistic.M18-M03  
**Significance threshold:** 0.02  
**Multiple testing correction:** Benjamini & Hochberg (FDR)

Selected results:

| Column              | Type           | Name                                                                             | Size | Score     | Adj. p-value | Median |
|---------------------|----------------|----------------------------------------------------------------------------------|------|-----------|--------------|--------|
| t.statistic.M03-D14 | KEGG name      | Glycosphingolipid biosynthesis - globo series                                    | 4    | 89.87E-2  | 12.92E-3     | 5.954  |
| t.statistic.M03-D14 | GOMF name      | glutathione peroxidase activity                                                  | 10   | 65.38E-2  | 16.68E-3     | 5.584  |
| t.statistic.M03-D14 | KEGG name      | Steroid biosynthesis                                                             | 7    | 74.16E-2  | 64.59E-4     | 4.479  |
| t.statistic.M18-M03 | KEGG name      | Rheumatoid arthritis                                                             | 15   | 82.46E-2  | 97.32E-8     | 3.948  |
| t.statistic.M18-M03 | GOCC slim name | immunoglobulin complex                                                           | 18   | 81.45E-2  | 15.15E-9     | 3.864  |
| t.statistic.M18-M03 | KEGG name      | Glycosphingolipid biosynthesis - globo series                                    | 4    | 87.72E-2  | 14.32E-3     | 3.734  |
| t.statistic.M18-M03 | GOMF name      | oxidoreductase activity, acting on the CH-CH group of donors, oxygen as acceptor | 6    | 86.33E-2  | 78.04E-4     | 3.719  |
| t.statistic.M18-M03 | KEGG name      | Other glycan degradation                                                         | 12   | 84.5E-2   | 83.32E-7     | 3.717  |
| t.statistic.M03-D14 | KEGG name      | Galactose metabolism                                                             | 18   | 46.27E-2  | 64.59E-4     | 3.666  |
| t.statistic.M18-M03 | KEGG name      | Glycosaminoglycan degradation                                                    | 12   | 77.86E-2  | 50.59E-6     | 3.606  |
| t.statistic.M03-D14 | KEGG name      | Arginine and proline metabolism                                                  | 31   | -31.86E-2 | 14.79E-3     | -3.485 |
| t.statistic.M03-D14 | KEGG name      | Protein digestion and absorption                                                 | 23   | -41.44E-2 | 60.1E-4      | -3.668 |
| t.statistic.M03-D14 | KEGG name      | Glycine, serine and threonine metabolism                                         | 16   | -43.93E-2 | 15.66E-3     | -4.282 |
| t.statistic.M03-D14 | GOMF name      | extracellular matrix structural constituent                                      | 22   | -55.54E-2 | 68.41E-5     | -4.485 |
| t.statistic.M18-M03 | KEGG name      | Mismatch repair                                                                  | 15   | -89.18E-2 | 15.27E-8     | -4.6   |
| t.statistic.M18-M03 | KEGG name      | DNA replication                                                                  | 28   | -90.61E-2 | 10.5E-15     | -4.998 |
| t.statistic.M03-D14 | KEGG name      | DNA replication                                                                  | 28   | -82.89E-2 | 46.14E-13    | -5.907 |
| t.statistic.M03-D14 | GOMF name      | glucosyltransferase activity                                                     | 8    | -82.83E-2 | 37.01E-4     | -8.1   |
| t.statistic.M03-D14 | GOMF name      | L-ascorbic acid binding                                                          | 11   | -82.33E-2 | 31.76E-5     | -8.431 |

Refer to Enrichment-table-13 for full results. Terms shown here are chosen automatically and are not necessarily the most relevant.

## 14. Remove imputed

Table 14

*Fold changes and p-values calculated with imputed data are kept.*

6286 protein groups

### Parameters:

Replace imputed values by [NaN].

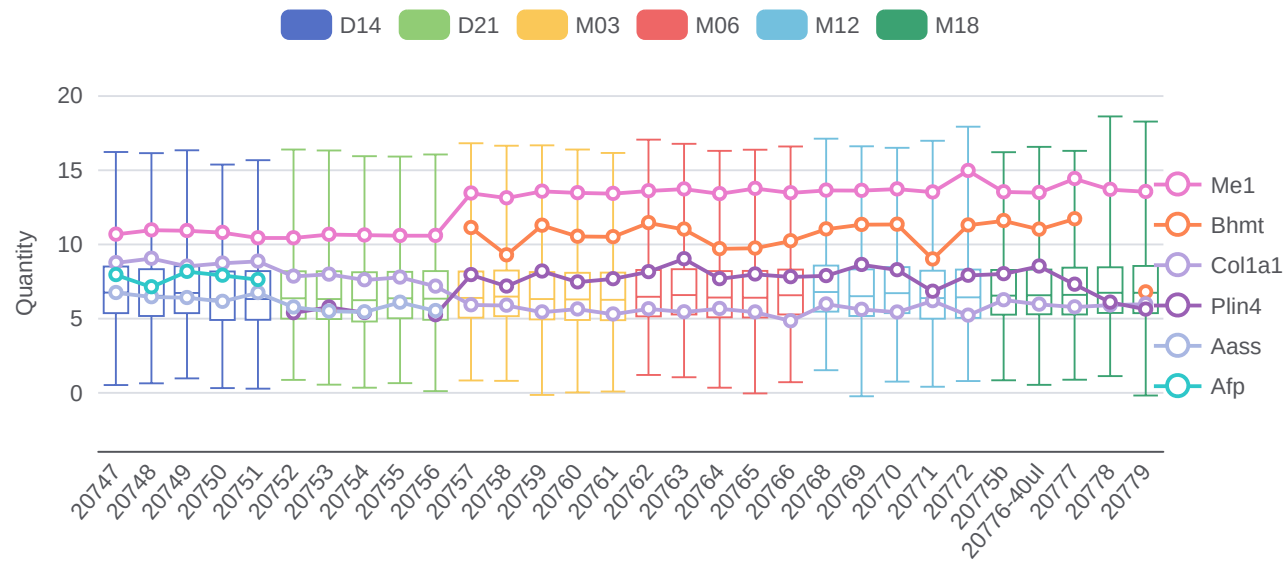
### Nr of replacements:

20757

Nr of protein groups with replacements: 2536

15. Boxplot

Overview of samples. Y-axis is usually in log2 scale.



Proteins significant in the M03-D14 comparison are shown.  
Samples 20778-79 appear like outliers in some respects

16. Rename headers

Add conditions to table headers.

Table 16

6286 protein groups

17. Order columns

Table cleanup.

Table 17

6286 protein groups

Parameters:  
Move default intensity columns [Quantity] first.